

CampusConnect — Database Schema

campus_connect_schema.sql

MySQL / MariaDB · Relational schema for the complaint & ticket management system

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1 -- =====
2 -- CampusConnect: Complaint & Ticket Management System
3 -- Relational Database Schema (MySQL / MariaDB syntax)
4 -- Designed from the existing frontend data model in js/store.js
5 -- =====
6
7 CREATE DATABASE IF NOT EXISTS campus_connect;
8 USE campus_connect;
9
10 -- -----
11 -- 1. USERS — base table for every account (student or admin)
12 -- -----
13 CREATE TABLE users (
14   user_id INT AUTO INCREMENT PRIMARY KEY,
15   name VARCHAR(100) NOT NULL,
16   email VARCHAR(150) NOT NULL UNIQUE,
17   password_hash VARCHAR(255) NOT NULL,
18   role ENUM('student', 'admin') NOT NULL,
19   phone VARCHAR(20),
20   created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
21 );
22
23 -- -----
24 -- 2. STUDENTS — 1:1 extension of users where role = 'student'
25 -- -----
26 CREATE TABLE students (
27   user_id INT PRIMARY KEY,
28   student_id VARCHAR(30) NOT NULL UNIQUE, -- e.g. CS-2023-042
29   department VARCHAR(100) NOT NULL,
30   FOREIGN KEY (user_id) REFERENCES users(user_id) ON DELETE CASCADE
31 );
32
33 -- -----
34 -- 3. ADMINS — 1:1 extension of users where role = 'admin'
35 -- -----
36 CREATE TABLE admins (
37   user_id INT PRIMARY KEY,
38   department VARCHAR(100) NOT NULL, -- e.g. Campus Grievance & Operations Cell
39   title VARCHAR(100), -- e.g. Chief Grievance Officer
40   FOREIGN KEY (user_id) REFERENCES users(user_id) ON DELETE CASCADE
41 );
42
43 -- -----
44 -- 4. CATEGORIES — lookup table (IT & Network Support, Hostel & Maintenance, ...)
45 -- -----
46 CREATE TABLE categories (
47   category_id INT AUTO INCREMENT PRIMARY KEY,
48   name VARCHAR(100) NOT NULL UNIQUE
49 );
50
51 -- -----
52 -- 5. TICKETS — the core grievance/ticket record
53 -- -----
54 CREATE TABLE tickets (
55   ticket_id VARCHAR(20) PRIMARY KEY, -- e.g. TKT-2026-101
56   title VARCHAR(200) NOT NULL,
57   description TEXT NOT NULL,
58   category_id INT NOT NULL,
59   priority ENUM('Low', 'Medium', 'High', 'Urgent') NOT NULL DEFAULT 'Medium',
60   status ENUM('open', 'in_progress', 'resolved', 'closed') NOT NULL DEFAULT 'open',
61   location VARCHAR(200),
62   submitted_by INT NOT NULL, -- FK to users (student who filed it)
63   assigned_to VARCHAR(150) DEFAULT 'Pending Assignment', -- staff/department name
64   created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
65   updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
66   FOREIGN KEY (category_id) REFERENCES categories(category_id),
67   FOREIGN KEY (submitted_by) REFERENCES users(user_id)
68 );
69
70 -- -----
71 -- 6. TICKET_TIMELINE — status history / audit trail per ticket
72 -- -----
73 CREATE TABLE ticket_timeline (
74   timeline_id INT AUTO INCREMENT PRIMARY KEY,
75   ticket_id VARCHAR(20) NOT NULL,
76   status ENUM('open', 'in_progress', 'resolved', 'closed') NOT NULL,
77   label VARCHAR(150) NOT NULL, -- e.g. "Assigned to Network Operations"
78   event_date DATETIME NOT NULL,
79   note TEXT,
80   FOREIGN KEY (ticket_id) REFERENCES tickets(ticket_id) ON DELETE CASCADE
81 );
82
83 -- -----
84 -- 7. TICKET_COMMENTS — conversation thread on a ticket
85 -- -----
86 CREATE TABLE ticket_comments (
87   comment_id INT AUTO INCREMENT PRIMARY KEY,
88   ticket_id VARCHAR(20) NOT NULL,
89   author_id INT NOT NULL, -- FK to users (student or admin who commented)
90   role ENUM('student', 'admin') NOT NULL,
91   comment_text TEXT NOT NULL,
92   comment_date DATETIME NOT NULL,
93   FOREIGN KEY (ticket_id) REFERENCES tickets(ticket_id) ON DELETE CASCADE,
94   FOREIGN KEY (author_id) REFERENCES users(user_id)
95 );
96
97 -- -----
98 -- Seed data: categories (matches the dropdowns in user-dashboard.html)
99 -- -----
100 INSERT INTO categories (name) VALUES
101 ('IT & Network Support'),
102 ('Hostel & Maintenance'),
103 ('Academics & Examination'),
104 ('Library & Digital Resources'),
105 ('Cafeteria & Mess'),
106 ('Accounts & Finance');
107
108 -- -----
109 -- Useful indexes for the queries the frontend already runs
110 -- (filter by student email, status, category, priority, search)
111 -- -----
112 CREATE INDEX idx_tickets_status ON tickets(status);
113 CREATE INDEX idx_tickets_priority ON tickets(priority);
114 CREATE INDEX idx_tickets_category ON tickets(category_id);
115 CREATE INDEX idx_tickets_submitted_by ON tickets(submitted_by);
```